

Assignment 3

Q1. Explain why the ReLU (Rectified Linear Unit) activation function is generally preferred over Sigmoid for the hidden layers of deep neural networks. Specifically, discuss how each handles the "vanishing gradient" problem during backpropagation.

Q2. You are training a deep network and observe that your Training Accuracy is 98% but your Validation Accuracy is only 72%.

- **A)** What is this phenomenon called?
- **B)** Name two specific techniques you could add to the architecture or training process to reduce this gap.

Q3. Consider a single fully connected (Dense) layer with 4 input neurons and 3 output neurons.

- Calculate the total number of learnable parameters (weights + biases) for this layer.
- *Hint: Don't forget the bias term for each output neuron.*

Q4. CNN Dimensions

You have an input image of size $32 \times 32 \times 3$ (RGB). You apply a Convolutional Layer with the following hyperparameters:

- **Filter (Kernel) Size:** 5×5 , **Stride:** 1, **Padding:** 0, Number of Filters: 10

Calculate the spatial dimensions (*Height* $\times$ *Width*) and the depth (channels) of the output feature map.

Q5. In a typical CNN architecture (like AlexNet), Max Pooling layers are inserted between Convolutional layers. What are the two primary benefits of doing this regarding computational cost and feature invariance?

Q6. Standard RNNs theoretically can remember long sequences, but in practice, they fail due to the Vanishing Gradient problem during Backpropagation Through Time (BPTT). Mathematically, what happens to the gradients as they are propagated back through many time steps (e.g., $t_{100} \rightarrow t_1$)?

Q7. You are training a model on a dataset with very limited computational resources (memory and training time) and a relatively small amount of training data.

- **A)** Would you choose an **LSTM** or a **GRU**?
- **B)** Justify your choice by comparing the **number of gates** and the resulting **parameter complexity** of both architectures. Which specific gate does the GRU combine/remove compared to the LSTM?

Q8. RNNs inherently understand the order of words because they process them one by one. Transformers process all words in parallel. Explain how a Transformer knows that "The cat ate the mouse" is different from "The mouse ate the cat" if it sees all words simultaneously.

Q9. In the context of Self-Attention, we compute a score using Query (Q), Key (K), and Value (V) vectors. If Q represents "what I am looking for" and K represents "the label of the content," what mathematical operation (discussed in class) is used to calculate the similarity between Q and K ? What happens to the V (content) if this similarity score is close to zero?

Q10. We discussed that BERT is an "**Encoder-only**" model, meaning it discards the Decoder component of the original Transformer. Please explain why?